

# 8259A Programmable Interrupt Controller (PIC)

## 1. Introduction

The **8259A PIC** acts as an interrupt manager between the CPU and multiple peripheral devices. It is essential for managing multiple hardware interrupts efficiently.

### Technical Detail

#### The Problem Without 8259A PIC:

- Only **one** hardware interrupt input (**INTR**) is available on the CPU.
- Multiple external devices must share the same INTR line.
- The CPU cannot determine:
  - Which device is requesting the interrupt?
  - What priority should be served first?
- There is no automatic generation of the interrupt vector number.

### Function / Operation

#### Available Solutions:

1. Using extra hardware or complex software to mask specific devices.
2. **Polling**: Checking all devices sequentially to find which requested service (slow).
3. **Using 8259A PIC**: Acting as a dedicated manager between the CPU and peripherals.

## 2. General Features

### Concept / Register

The 8259A PIC adds **8 vectored priority encoded interrupts** to the microprocessor.

### Technical Detail

#### Expansion Capability:

- Can be expanded to accept up to **64 interrupt requests**.
- Expansion requires:
  1. One **Master** 8259A.
  2. Eight **Slave** 8259As.

### 3. Internal Registers and Priority

#### Concept / Register

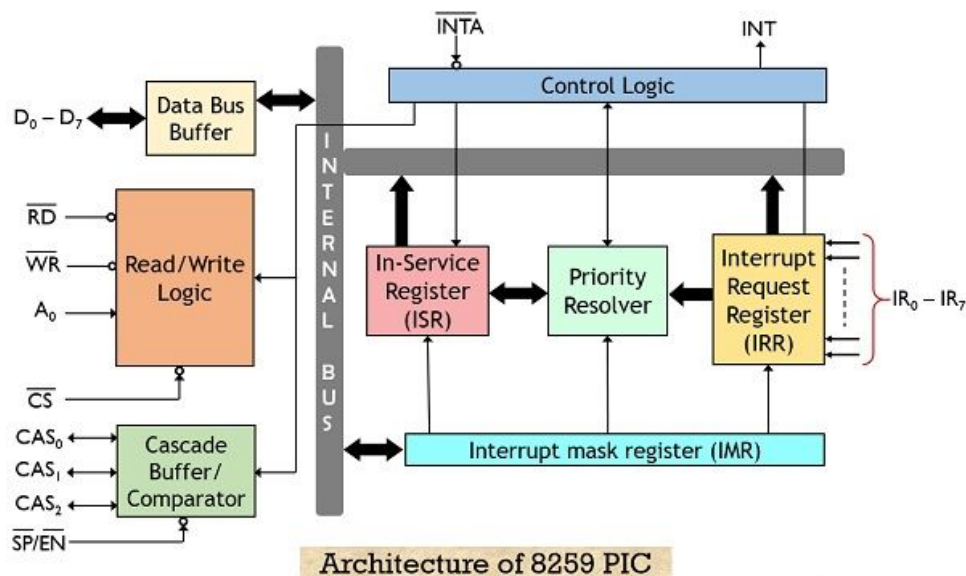
##### Functional Registers:

- **IRR (Interrupt Request Register):** Stores interrupt requests generated by peripheral devices.
- **ISR (In-Service Register):** Stores interrupts which are currently being executed by the processor.
- **IMR (Interrupt Mask Register):** Allows individual masking of IR lines via software.

#### Function / Operation

##### Interrupt Request Lines ( $IR_0-IR_7$ ):

- Receive interrupt requests from peripherals or slaves.
- **Priority:** Default is  $IR_0$  (highest) to  $IR_7$  (lowest).
- **Flexibility:** Priority levels can be rotated via software.



Electronics Desk

Figure 1: 8259A Internal Block Diagram

## 4. Pin Descriptions and Functions

|                 |    |      |                    |
|-----------------|----|------|--------------------|
| $\overline{CS}$ | 1  | 28   | V <sub>CC</sub>    |
| $\overline{WR}$ | 2  | 27   | A0                 |
| $\overline{RD}$ | 3  | 26   | $\overline{INTA}$  |
| D7              | 4  | 25   | IR7                |
| D6              | 5  | 24   | IR6                |
| D5              | 6  | 23   | IR5                |
| D4              | 7  | 8259 | IR4                |
| D3              | 8  | PIC  | IR3                |
| D2              | 9  | 20   | IR2                |
| D1              | 10 | 19   | IR1                |
| D0              | 11 | 18   | IR0                |
| CAS0            | 12 | 17   | INT                |
| CAS1            | 13 | 16   | $\overline{SP/EN}$ |
| gnd             | 14 | 15   | CAS2               |

Figure 2: 8259A Pin Diagram

### 4.1 Interface Pins

#### Technical Detail

- **INT (Interrupt):**
  - **Master:** Connects to the CPU's **INTR** pin.
  - **Slave:** Connects to a **Master IR** pin.
- **$\overline{INTA}$  (Interrupt Acknowledge):** Connects to the system's **INTA** signal. **Only the Master's  $\overline{INTA}$  signal is connected.**
- **$D_0$ – $D_7$ :** Bidirectional data bus lines for exchanging command words, status, mask bits, and interrupt type numbers with the CPU.

## 4.2 Control and Cascading Pins

### Function / Operation

- $A_0$  (**Address Line**): Selects which internal register is accessed:
  - **0**: Command Register.
  - **1**: Data Register.
- $\overline{SP}/\overline{EN}$  (**Slave Program / Enable Buffer**):
  - **Buffered Mode**: Controls data bus transceivers.
  - **Non-Buffered Mode**: Programs the device as **Master (1)** or **Slave (0)**.
- $CAS_0-CAS_2$ : Used for cascading multiple PICs. Allows comparison of IDs to identify which slave issued the interrupt.
  - Master's CAS pins (**Outputs**) are connected in parallel to Slave CAS pins (**Inputs**).

## 5. Interrupt Acknowledge Pulse Sequence

When an interrupt occurs, the CPU sends **two acknowledge pulses**:

1. **1st Pulse**: The 8259A identifies which device caused the interrupt.
2. **2nd Pulse**: The 8259A places the **interrupt vector number** onto the data bus for the CPU to read.